

Kisan ISP - External Payment Public API Guide

Production Documentation & Integration Manual | Version 1.0.0

1. Overview & Architecture

The External Payment Gateway (externalpayment) provides an enterprise REST API that allows third-party payment gateways, bank channels, kiosk aggregators, and custom portals to push instant subscriber recharges, query customer balances, and synchronize billing transactions automatically with Kisan ISP.

It mirrors 100% of the proven architecture from existing payment integrations (such as eSewa) while introducing direct push recharge capabilities where an external service can recharge an account directly using a customer username and duration without requiring a prior inquiry request.

2. Endpoints Summary

POST /api/externalpayment/access-token	Obtains OAuth2 Bearer access token using username & password
POST /api/externalpayment/login	Direct JSON login alternative returning Bearer token
GET /api/externalpayment/inquiry/:id	Search user by PPPoE username, customer ID, phone, or email
GET /api/externalpayment/user/:username	Lookup customer directly by PPPoE username (e.g. karnkalyan)
POST /api/externalpayment/payment	Direct Push Recharge: renews subscription, RADIUS & invoice
POST /api/externalpayment/recharge	Alias to /payment for direct recharge push
POST /api/externalpayment/status	Transaction reconciliation & status query by transaction_code
GET /api/externalpayment/payment-modes	List available payment methods (EXTERNAL, CASH, ONLINE, etc.)
GET /externalpayment	Frontend management dashboard & instant recharge tester

3. Authentication Methods

The API supports multiple authentication strategies to ensure seamless compatibility with external servers, cron jobs, and webhooks:

Bearer Token	Header: Authorization: Bearer <access_token> (expires in 24 hours)
HTTP Basic Auth	Header: Authorization: Basic <base64(username:password)>
Payload Credentials	Body: "auth_username" & "auth_password" (or "username" & "password")
API Key	Header: x-api-key: <api_key>

External Payment API - Customer Inquiry & Recharge

Step-by-Step API Specification & Direct Recharge Workflows

4. Customer Inquiry (Inquiry API)

Query customer status, current subscribed package, plan expiration date, and all available online packages for renewal. The inquiry parameter accepts: PPPoE username (e.g. karnkalyan), Customer Unique ID (e.g. CUST-1001), mobile phone number, or email.

HTTP Request:

```
GET https://cms.kisan.net.np/api/externalpayment/inquiry/karnkalyan
Authorization: Basic ZXh0ZXJyYXxfXNwXzE6RXh0ZXJyYXxASVNQIyExMjAyNQ==
```

Success Response (JSON):

```
{
  "response_code": 0,
  "response_message": "success",
  "customer": {
    "id": 142,
    "customer_unique_id": "CUST-1001",
    "customer_name": "Kalyan Karn",
    "username": "karnkalyan",
    "phone": "9800000000",
    "expiry_date": "2026-10-01",
    "status": "active",
    "is_rechargeable": true
  },
  "current_package": {
    "id": 12,
    "name": "KISAN-75-1-TV",
    "duration": "1 month",
    "amount": 1200
  },
  "packages": [
    { "id": 12, "name": "KISAN-75-1-TV", "duration": "1 month", "amount": 1200 },
    { "id": 13, "name": "KISAN-75-3-TV", "duration": "3 months", "amount": 3400 }
  ]
}
```

5. Direct Push Recharge API

The primary endpoint for pushing payment confirmation and instant recharge. When called, the system automatically resolves the customer, calculates the new expiration date, provisions RADIUS, syncs the sales invoice to accounting, and marks the account active.

HTTP Request (cURL):

```
curl -X POST "https://cms.kisan.net.np/api/externalpayment/payment" \
-H "Content-Type: application/json" \
-u "external_isp_1:External@ISP#1!2025" \
-d '{
  "username": "karnkalyan",
  "payment_mode": "EXTERNAL",
  "duration": "1 month",
  "amount": 1200,
  "transaction_code": "EXT-TXN-9847291"
}'
```

Success Response (JSON):

```
{
  "response_code": 0,
  "response_message": "Payment and recharge successful",
  "data": {
    "transaction_code": "EXT-TXN-9847291",
    "reference_code": "EXT-20260916-B97FC73659FE-24",
    "order_id": 24,
    "customer_id": "CUST-1001",
    "customer_name": "Kalyan Karn",
    "username": "karnkalyan",
    "payment_mode": "EXTERNAL",
    "amount": 1200,
    "package_name": "KISAN-75-1-TV",
    "package_duration": "1 month",
    "new_expiry_date": "2026-11-01T18:15:00.000Z",
    "radius_provisioned": true
  }
}
```

RADIUS Provisioning, Accounting Integration & Reconciliation

Expiration Attribute	Updated in radreply table to the exact new calculated package end date.
Session Disconnection	Sends a disconnect request (CoA / PoD) to NAS/BRAS so the user immediately reconnects with the renewed speed profile and updated quota.
Service Status	Activates connection user status and sets customer service connection to active.

Alternative URL	https://cms.kisan.net.np/services/externalpayment
Sidebar Menu	Services -> 3rd Party Services -> External Payment
Live Tester Tab	Allows operators to type username & duration to test live push recharges directly from the web interface.